

Trigonometry applied to solids of revolution

Cambridge insert: the axial section is a plane triangle, and every cone problem lives inside it.

LESSON 43–44

2 × 40 MIN

GEOMETRY 11, §19 (EXTENSION)

IGCSE E6.6

LESSON CLOCK

14'

24'

24'

22'

20'

6'

The axial section	14 min
The semi-vertical angle	24 min
Frustums and inscribed solids	24 min
Combined problems	22 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Extract the axial section of a cone or a frustum and solve it.
- Find the semi-vertical angle of a cone.
- Use the sine and cosine rules inside a solid of revolution.
- Combine a trigonometric result with a volume formula.

TEXTBOOK

National	pp. 217–222
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.6

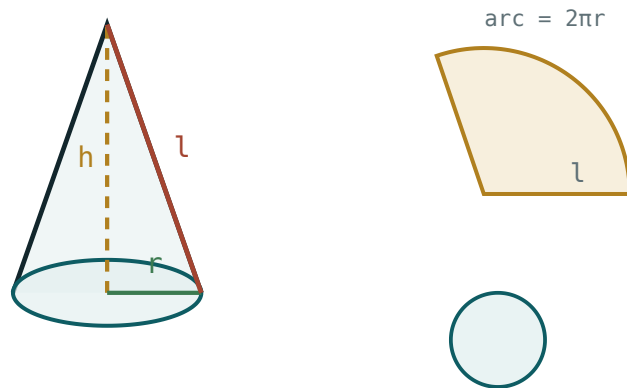
01

Explanation

The axial section

THE ONE MOVE

Cut the solid through its axis. A cone gives an isosceles triangle of base $2r$ and height h ; a cylinder gives a rectangle $2r$ by h ; a frustum gives a trapezium. Draw that flat figure separately and solve it as a plane problem.



$$S = \pi r^2 + \pi r l - \text{a circle and a sector}$$

The right triangle inside the cone: r , h and l .

Every 3-D question about a solid of revolution reduces to this figure, exactly as every cuboid question in Grade 10 reduced to one extracted triangle.

The semi-vertical angle

The angle α between the axis and a slant line is the **semi-vertical angle**; the full angle at the apex is 2α .

$$\tan \alpha = \frac{r}{h} \quad \sin \alpha = \frac{r}{l} \quad \cos \alpha = \frac{h}{l}$$

The angle β the slant makes with the **base** is the complement: $\beta = 90^\circ - \alpha$, with $\tan \beta = \frac{h}{r}$.

TWO DIFFERENT ANGLES

A question asking for “the angle at the vertex” wants 2α ; one asking for “the angle of the slant with the base” wants β . Read which, and mark it on the extracted triangle.

Frustums and inscribed solids

The axial section of a frustum is a trapezium; drop a perpendicular from the top corner to make a right triangle with legs h and $R - r$ and hypotenuse l . Then

$$\tan \beta = \frac{h}{R - r}$$

Inscribed cone in a sphere. The axial section is an isosceles triangle inscribed in a circle. With base radius r and height h in a sphere of radius R :

$$r^2 = h(2R - h) \text{ (the intersecting-chord theorem)}$$

Cylinder in a cone. Similar triangles in the axial section relate the cylinder's radius to its height, and the optimisation of Grade 11 Quarter I then finds the largest one.

Combined problems

A full problem chains three steps:

1. Extract and label the axial section.
2. Solve it with right-angle trigonometry, or the sine or cosine rule.
3. Feed the length found into the volume or surface-area formula.

Example. A cone has slant height 20 and semi-vertical angle 35° . Then $r = 20 \sin 35^\circ \approx 11.47$ and $h = 20 \cos 35^\circ \approx 16.38$, so $V = \frac{1}{3}\pi(11.47)(16.38) \approx 2258$ and $S = \pi r(r + l) \approx 1134$.

02 Worked examples

EXAMPLE 1 A cone has radius 9 and height 12. Find the semi-vertical angle and the angle of the slant with the base.

$$\textcircled{1} \quad \tan \alpha = \frac{9}{12} = 0.75$$

$$\textcircled{2} \quad \alpha \approx 36.9^\circ$$

Semi-vertical.

$$\textcircled{3} \quad \beta = 90 - 36.9 = 53.1^\circ$$

With the base.

ANSWER

$$\alpha \approx 36.9^\circ, \beta \approx 53.1^\circ$$

EXAMPLE 2 A cone has slant height 20 and semi-vertical angle 35° . Find its volume.

$$① \quad r = 20 \sin 35^\circ \approx 11.47$$

$$② \quad h = 20 \cos 35^\circ \approx 16.38$$

$$③ \quad V = \frac{1}{3} \pi (11.47)(16.38)$$

$$④ \quad \approx 2258$$

ANSWER

≈ 2260 cubic units

EXAMPLE 3 A frustum has radii 10 and 4 and height 8. Find the angle the slant makes with the base.

$$① \quad R - r = 6$$

$$② \quad \tan \beta = \frac{8}{6}$$

$$③ \quad \beta \approx 53.1^\circ$$

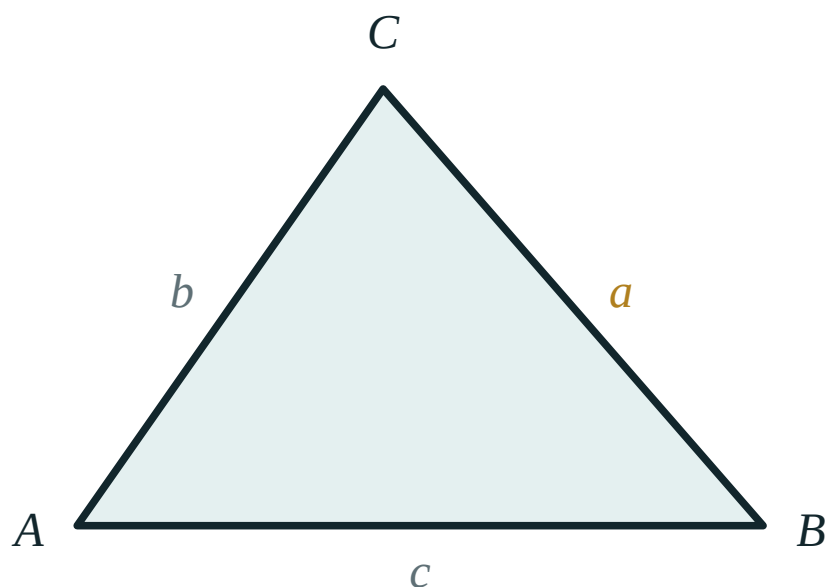
ANSWER

$\approx 53.1^\circ$

03 Interactive model

Cut a cone of plasticine through its axis and hold up the triangle.

The axial triangle



b 7

c 9

A 55°

a 7.6

B 49° C 76°

area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Axial section	O'qli kesim	Осевое сечение
Semi-vertical angle	Yarim uch burchagi	Половина угла при вершине
Slant height	Yasovchi	Образующая
Solid of revolution	Aylanish jismi	Тело вращения
Angle of the slant with the base	Yasovchi va asos burchagi	Угол образующей с основанием
Inscribed cone	Ichki chizilgan konus	Вписанный конус
Extracted triangle	Ajratilgan uchburchak	Выделенный треугольник
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

The axial section of a cone is:

The axial section of a frustum is:

$\tan \alpha$ for the semi-vertical angle:

\frac{h}{r}
"/>

\frac{r}{h}
"/>

\frac{r}{l}
"/>

\frac{h}{l}
"/>

The angle at the apex is:

$$\alpha$$

$$2\alpha$$

$$90 - \alpha$$

$$180 - \alpha$$

A cone inscribed in a sphere satisfies:

$$r^2 = h(2R - h)$$

$$r = R$$

$$h = R$$

$$r^2 = R^2 + h^2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Semi-vertical angle with $r = 3$, $h = 4$

ANSWER $\approx 36.9^\circ$

2. Semi-vertical angle with $r = 5$, $h = 12$

ANSWER $\approx 22.6^\circ$

3. Angle with the base for $r = 3$, $h = 4$

ANSWER $\approx 53.1^\circ$

4. r from $l = 10$, $\alpha = 30^\circ$

ANSWER 5

5. h from $l = 10$, $\alpha = 30^\circ$

ANSWER $5\sqrt{3}$

6. Axial section of a cylinder

ANSWER a rectangle

7. Axial section of a frustum

ANSWER a trapezium

Medium · the standard question

7 PROBLEMS

1. $r = 9, h = 12$: both angles

ANSWER 36.9° and 53.1°

2. $l = 20, \alpha = 35^\circ$: the volume

ANSWER ≈ 2260

3. Same: the total surface area

ANSWER ≈ 1130

4. Frustum $R = 10, r = 4, h = 8$: the slant angle with the base

ANSWER $\approx 53.1^\circ$

5. Same: the slant height

ANSWER 10

6. A cone with apex angle 60° and $h = 12$: its radius

ANSWER $4\sqrt{3} \approx 6.93$

7. A cone of $r = 6, h = 8$ inscribed in a sphere: the sphere's radius

ANSWER 6.25

Hard · stretch and reason

7 PROBLEMS

1. A cone with semi-vertical angle 30° and volume 1000: its height

ANSWER ≈ 13.1

2. A cylinder of greatest volume inscribed in a cone of $r = 6$, $h = 12$: its dimensions

ANSWER $r = 4$, $h = 4$

3. A cone inscribed in a sphere of radius 10 with height 16: its base radius

ANSWER 8

4. A frustum of slant height 13 and radii 12 and 7: its height and volume

ANSWER 12, ≈ 3630

5. A cone of maximum volume in a sphere of radius R : its height

ANSWER $\frac{4R}{3}$

6. Two cones with the same slant height 15 and angles 20° , 40° : the volume ratio

ANSWER $\approx 1 : 3.5$

7. Prove $r^2 = h(2R - h)$ for a cone in a sphere

ANSWER Intersecting chords through the axis

07 Homework

Homework — 5 tasks

Draw the axial section separately, and mark the angle asked for.

1. A cone has radius 7 and height 24. Find the semi-vertical angle, the apex angle and the angle of the slant with the base.
2. A cone has slant height 26 and semi-vertical angle 40° . Find its radius, height and volume.
3. A frustum has radii 15 and 9 and height 8. Find its slant height and the angle with the base.
4. A cone of base radius 9 and height 12 is inscribed in a sphere. Find the sphere's radius.

5. Explain in three sentences why the axial section reduces a 3-D problem to a 2-D one.